



 AgroSmart

Agro-Smart

IoT-Based Smart Agriculture
Monitoring System

1st Progress Presentation

KAHNDSE251F-005
KAHNDSE251F-007
KAHNDSE251F-009
KAHNDSE251F-032

Introduction to Smart Agriculture Monitoring

- Agriculture heavily depends on environmental conditions such as temperature, humidity, and soil moisture.
- Traditional farming relies on manual observation, which is inefficient and inaccurate.
- IoT enables real-time monitoring and automation in agriculture.
- AgroSmart combines IoT sensors, cloud computing, and data analytics.



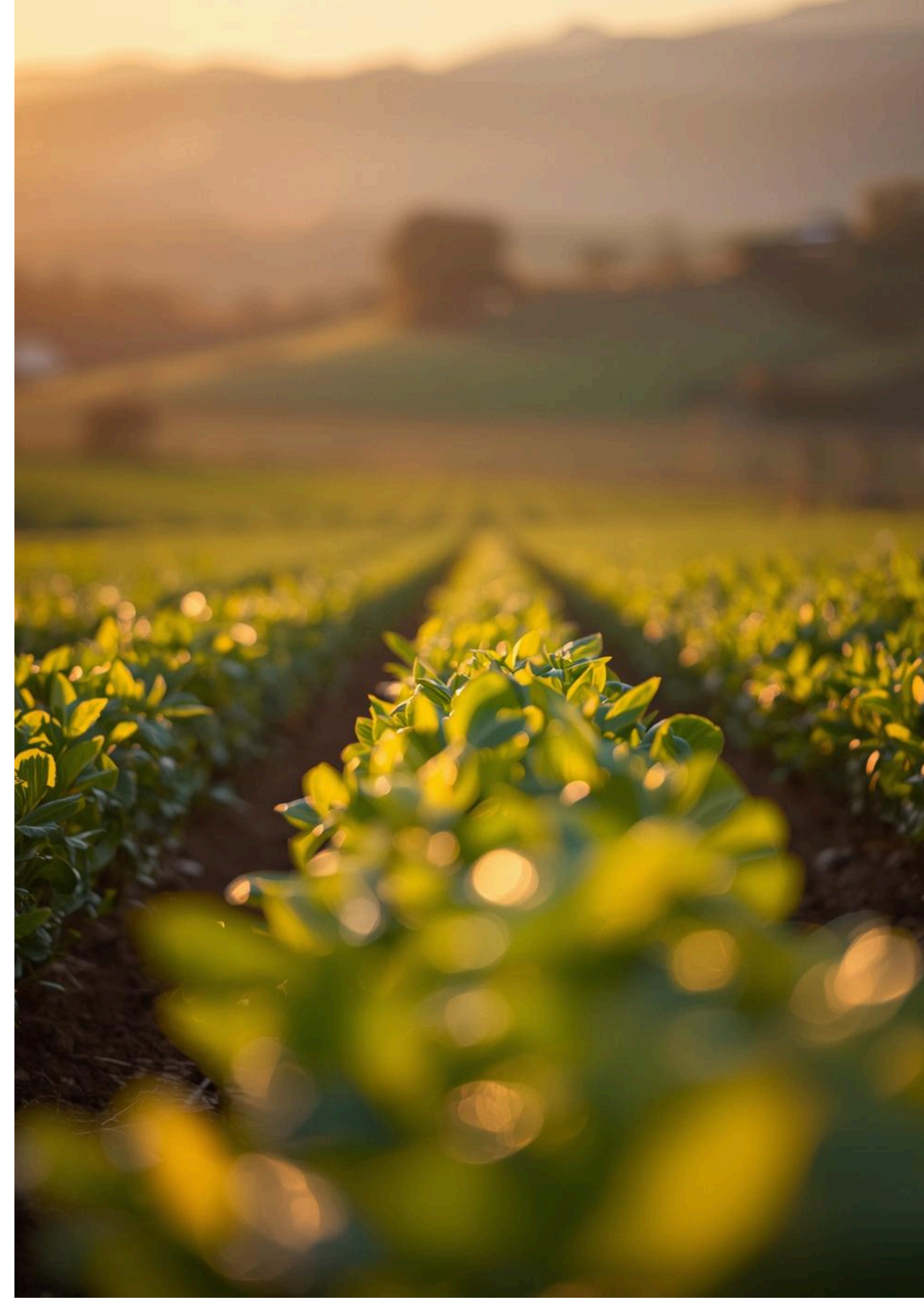
Aims & Objectives

Aim of the Project

- The aim of the AgroSmart project is to develop an IoT-based smart agriculture monitoring system that collects real-time environmental data and uses data analysis and AI/ML techniques to provide actionable insights for improving irrigation efficiency and crop suitability.

Objectives of the Project

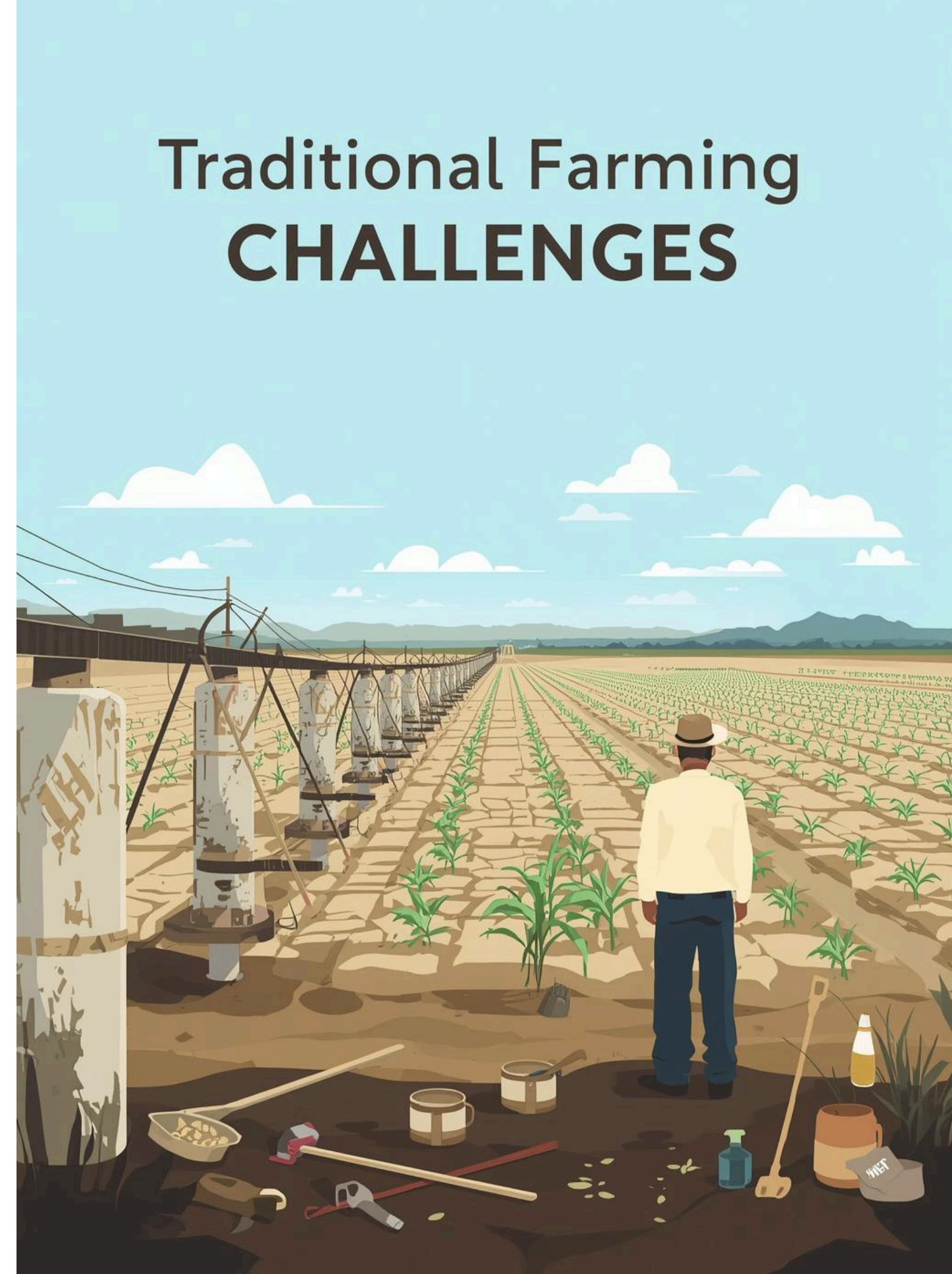
- To design and implement an IoT-based agriculture monitoring system using sensors and ESP32.
- To collect real-time data such as temperature, humidity, soil moisture, and light intensity.
- To store and visualize sensor data using a dashboard and a mobile application
- To analyze historical datasets and live sensor data to identify patterns and trends.
- To provide smart irrigation recommendations and crop-related insights based on analyzed data.



Problem Statement

- Lack of real-time monitoring in traditional farming.
- Inefficient water usage due to manual irrigation.
- Difficulty in analyzing environmental trends.
- Farmers do not have easy access to data-driven insights.
- Limited use of technology in small-scale farming.

Traditional Farming **CHALLENGES**



Methodology Overview

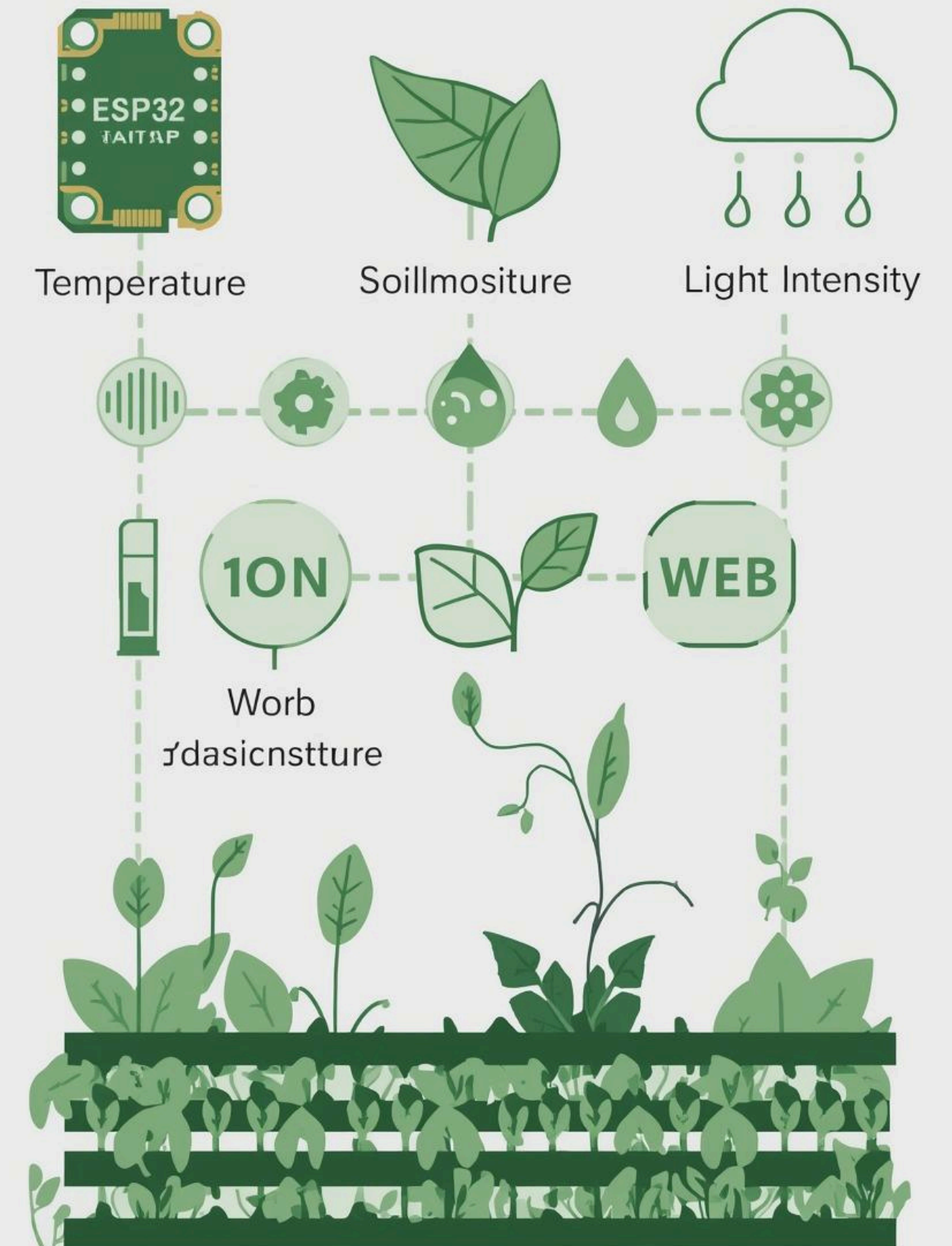
Hardware Layer

- ESP32 microcontroller
- DHT11 – Temperature & Humidity
- Capacitive Soil Moisture Sensor
- LDR – Light Intensity
- Relay Module – Smart Irrigation

Software & Cloud Layer

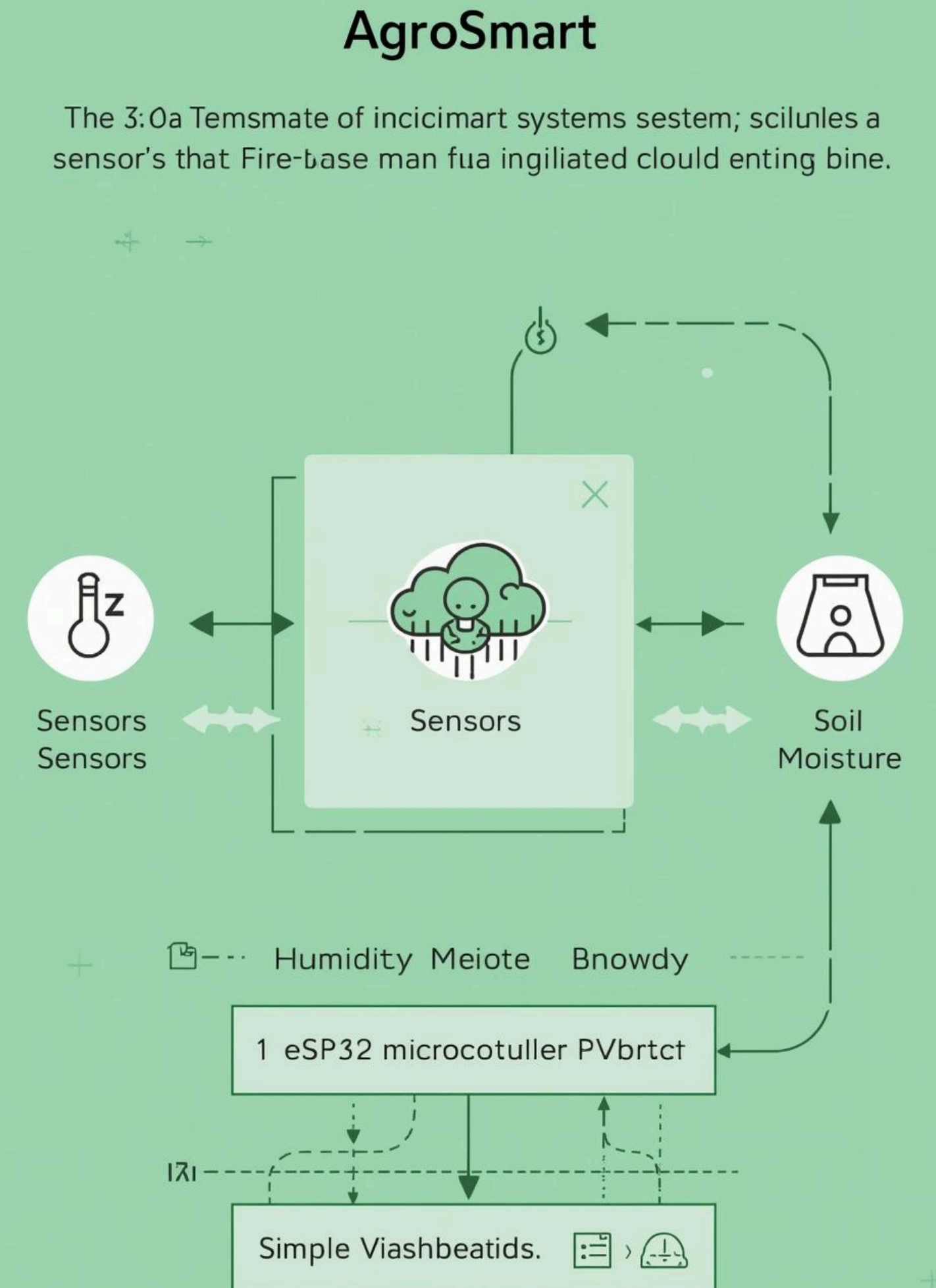
- Firebase Realtime Database
- Web-based Dashboard
- Data transmission via WiFi

IoT-based agricultural monitoring system



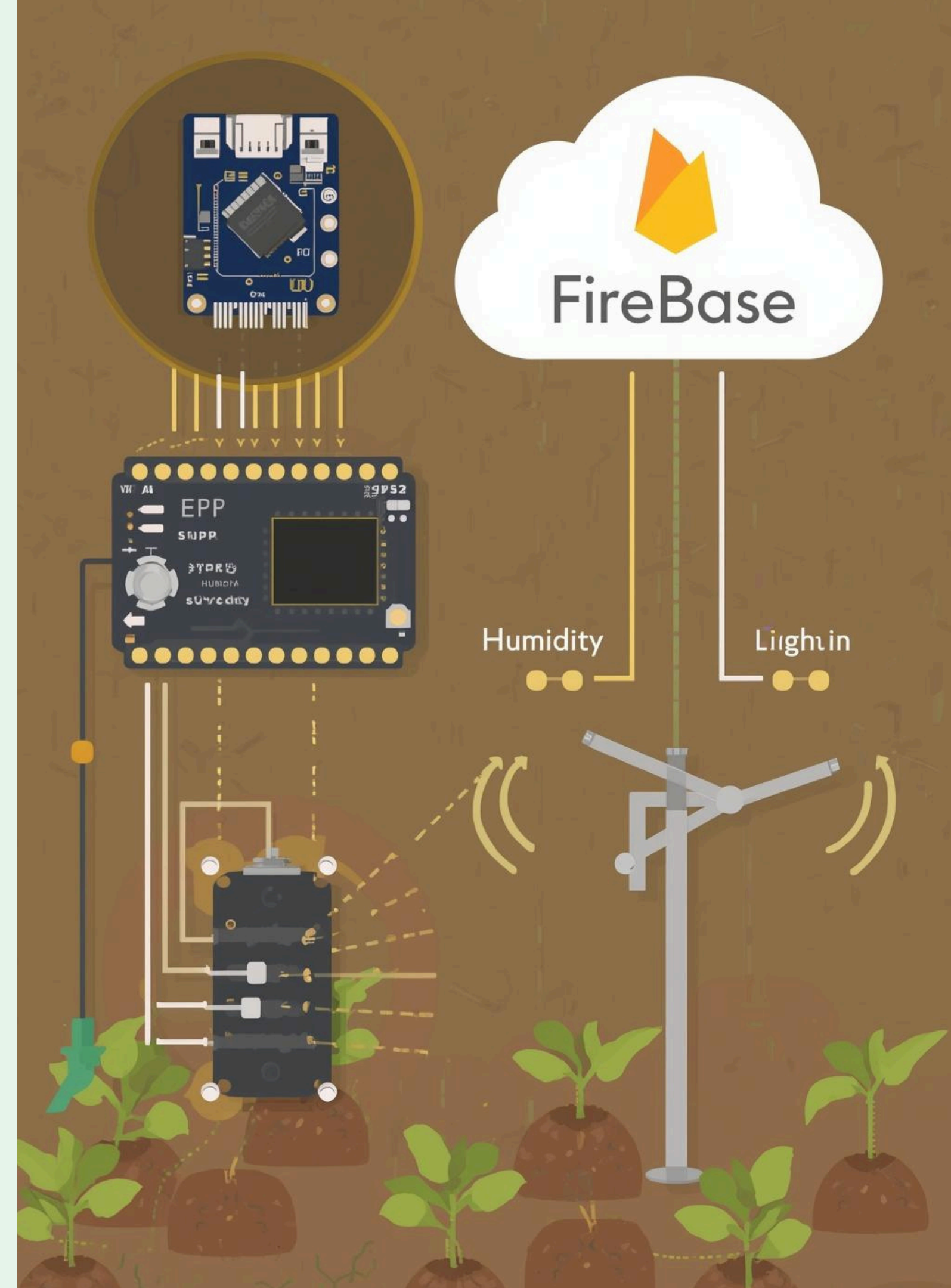
System Workflow

1. Sensors collect real-time environmental data.
2. ESP32 processes the sensor readings.
3. Data is sent to Firebase Realtime Database.
4. Dashboard displays live sensor data.
5. Smart irrigation is controlled using soil moisture levels.



Current Work Completed

- Hardware setup completed.
- Sensors tested and calibrated.
- ESP32 successfully connected to WiFi.
- Real-time data uploaded to Firebase.
- Smart irrigation logic implemented.
- Basic web dashboard developed.



Thank you!

